

AI WeatherWise API

Project title: “ AI WeatherWise Api ”

Team Leader: G.Kaviya

Team Members:

- M.Keerthana
- S.Janani
- J.Janapriya
- D.Jenifer

2: Requirement Analysis Phase

Introduction:

The Requirement Analysis Phase focuses on identifying the hardware, software, programming languages, libraries, database, APIs, and other resources required to develop AI-WeatherWise. In this phase, the technical and functional requirements of the system are analyzed to ensure that the project can be developed and implemented effectively.

Functional Requirements:

The functional requirements of AI-WeatherWise include:

- User registration and login.
- User authentication and authorization.
- Retrieval of weather information through APIs.
- Display of weather-related information to users.
- Storage and management of required application data.
- AI-based interaction or analysis using the integrated AI service. □ Handling of user requests and displaying appropriate responses.
- Secure management of user information.

Non-Functional Requirements:

The non-functional requirements include:

- **Usability** – The application should be simple and easy to use.
- **Performance** – Weather information and user requests should be processed efficiently.
- **Security** – User credentials and sensitive information should be protected.
- **Reliability** – The system should provide consistent responses when required services are available.
- **Scalability** – The application should allow additional features and modules to be added in the future.
- **Maintainability** – The code should be organized so that future modifications can be made easily.

Technical Requirements:

The proposed system uses **Node.js and Express.js** for backend development and **MongoDB with Mongoose** for database management.

JavaScript is used as the primary programming language.

Google Gemini AI is integrated to generate intelligent weather summaries and personalized recommendations.

JWT and bcrypt are used for authentication and password security.

Postman or Thunder Client is used for API testing, and **Visual Studio Code** is used for development.

OpenWeatherMap provides real-time weather data, with **JSON** used for data exchange.

Database:

MongoDB Atlas is used as the database platform for AI-WeatherWise. It provides cloudbased database storage for maintaining the required application data. **Mongoose** is used to establish communication between the Node.js application and MongoDB.

The database can be used to manage information such as:

- User details
- Authentication-related information
- Application data
- Weather-related data where required

API Requirements:

AI-WeatherWise uses external APIs to obtain weather-related information and provide data to the application. API integration enables the system to retrieve current weather information and other required weather data.

The API-based architecture allows the application to receive updated weather information instead of depending only on manually stored data.

Conclusion:

The Requirement Analysis Phase identifies the technical and functional requirements needed to develop AI-WeatherWise. By defining the hardware, software, programming languages, libraries, database, APIs, and functional requirements in advance, the development process can be carried out in a structured manner. These requirements provide the foundation for the next phase, which is the Project Design Phase.